RX5
Care Monitoring System



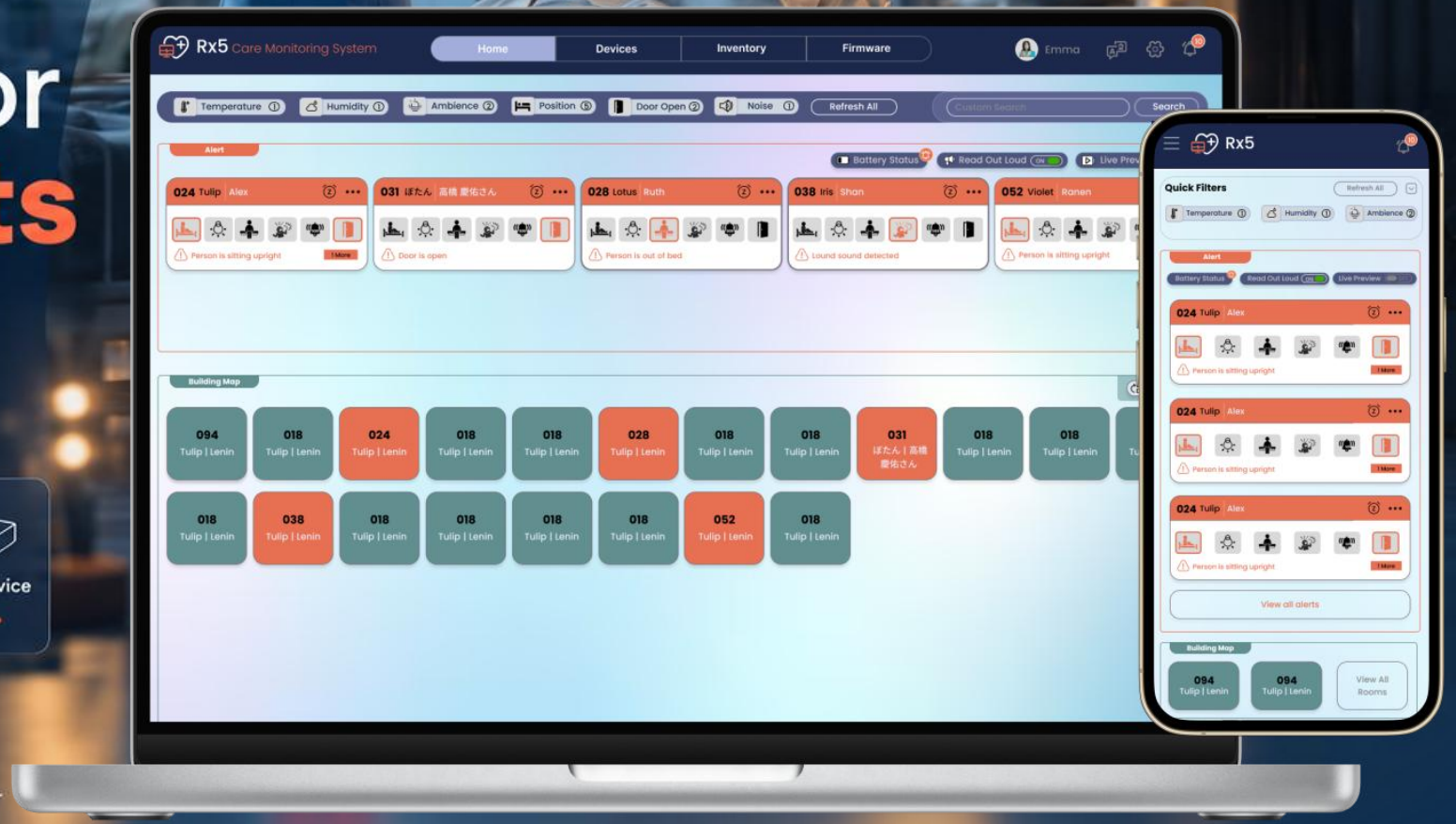
Best Design Award
Japan

UX / PRODUCT DESIGN CASE STUDY

Smarter Care for Elderly Patients

A real-time monitoring system that connects multiple sensors and devices to help trained nurses ensure the safety and well-being of elderly patients.

- Camera
- Floor Sensor
- Door Sensor
- Call Button
- Temperature Sensor
- RX5 Device



Primary Users
Trained Nurses

Domain
Elderly Care / Hospital

My Role
Lead Product Designer

One connected view of elderly care operations

Product overview • Web and mobile experience

RX5 Care Monitoring System is a connected care platform that helps nurses and care teams monitor patient-related signals and room conditions from one shared interface. It brings together sensor readings and alerts for temperature, humidity, ambience, patient position, door status, noise, and SOS-related events—making it easier to spot changes and identify where attention may be needed.

The web app supports wider operational oversight through the Home dashboard, building map, patient details, device registry, inventory, room and nursing-station setup, device configuration, and firmware management. The mobile app brings key workflows into a compact, responsive experience with quick filters, alert cards, navigation, device records, and firmware status. Together, both experiences connect monitoring signals with the room, patient, and device context teams need for follow-up.

CORE VALUE

Enabling safer, more proactive elderly care through real-time monitoring, intelligent alerts, and connected care technology.

Recognition - Award

Awarded in Japan for outstanding design and a simple, intuitive workflow that is easy to understand and use.

DESIGN INTENT

Turn a complex ecosystem of sensors, devices, rooms, and alerts into a coherent operational experience—without altering the underlying product screens.

The problem space

Care teams need to interpret signals from multiple connected sources while managing the devices that collect and relay them.

Fragmented signals

Temperature, humidity, ambience, position, door state, noise, and SOS signals each communicate a different kind of condition.

Time-sensitive attention

Alerts such as a patient being out of bed, a door being open, or loud sound being detected need to be visible without searching through multiple screens.

Operational complexity

Devices, room assignments, nursing stations, extensions, firmware versions, and update states create a second layer of administrative work.

- The experience must serve two connected jobs: monitor care conditions and maintain the connected-device ecosystem.
- A useful interface needs both a fast-glance operational view and deeper detail for investigation, configuration, and maintenance.

Lead Product Designer

A product-level contribution spanning problem framing, information architecture, responsive experience design, interface definition by collaboration with hardware team, product managers and stakeholder.

Product framing

Clarified the product story around monitoring, response, device administration, and system maintenance.

UX architecture

Organized the experience into Home, Devices, Inventory, and Firmware, with patient details and inventory sub-sections.

Interaction design

Defined scan patterns for alerts, sensor indicators, status states, tables, configuration forms, and easy navigation.

Cross-platform design

Connected web and mobile workflows while adapting density, navigation, and card layout to each form factor.

- Owned the end-to-end design narrative presented in this case study, from product context to interface decisions and reflection.
- Worked across system-level workflows with hardware team and individual screen states—not only visual polish.
- Collaboration areas to document with actual project evidence: engineering feasibility, product prioritization, stakeholder reviews, and validation feedback.

Discovery: understand the work behind the screens

The visible product surfaces point to a workflow that moves from noticing a signal to identifying a room or patient, investigating context, and maintaining the connected system.

Monitor

Scan alerts and sensor categories; recognize which rooms or patients need attention.

Investigate

Open a patient detail view to review identifiers, sensor states, critical ambience, door status, SOS, and activity comments.

Administer

Assign devices to rooms and nursing stations; add devices manually or through QR scanning.

Maintain

Review hardware/software versions, compare available firmware, and trigger update actions.

- Discovery synthesis from the supplied screens: the product serves clinical monitoring and technical administration in one ecosystem.
- Design implication: preserve a clear separation between urgent operational signals and lower-frequency setup/maintenance tasks.

Strategy: make the right information visible at the right depth

The interface uses layered disclosure: a fast overview for awareness, structured lists for management, and detail/configuration views for deliberate action.

1. Scan first

Use alert cards, status colors, sensor icons, and a building map to establish a rapid operational overview.

2. Add context

Connect room/patient identifiers with sensor readings and linked device information so a signal is interpretable.

3. Act in place

Provide relevant actions near the object being managed: patient detail tabs, device configuration, add-device flows, and firmware update controls.

- Prioritize clarity of status and location before exposing secondary metadata.
- Use consistent navigation and component patterns to help users transfer knowledge between monitoring and administration.

Information architecture: four primary workspaces

The top-level navigation groups tasks by user intent and separates live operations from system administration.

HOME

Alert overview, sensor quick filters, refresh/search controls, patient cards, optional live preview, and building map.

DEVICES

Device registry with base/mobile/door/floor sensors, captured data, last-updated dates, status, configurations and actions.

INVENTORY

Assigned device list, device-type filters, configuration, add-device flow, rooms, nursing stations, and mobile extensions.

FIRMWARE

Filter by all/RX device/app; compare hardware/software versions, available version, last updated, and update actions.

- Patient detail provides a focused drill-down with Details, Live, Alerts, sensors assigned and Settings tabs.
- Inventory is subdivided by Device List, Rooms & Nursing Stations, and Mobile Extension to keep related administration tasks together.
- This architecture gives the dashboard a clear operational role while keeping setup and maintenance workflows accessible but distinct.

End-to-end workflow: from signal to action

A coherent monitoring journey should support the next meaningful decision at each step.

01 • Detect

A sensor state or alert card surfaces a condition such as sitting upright, out of bed, door open, loud sound, ambience issue and more

02 • Locate

Use patient/room identifiers and the building map to understand where the condition is associated.

03 • Investigate

Review the alert details, visit the patient, identify the root cause of the issue, and take the necessary steps to resolve it.

04 • Maintain

For infrastructure issues, check device assignments and configurations in Devices or Inventory, and monitor all patients.

- Design opportunity: keep alert meaning, patient/room identity, and the next action visually connected.
- Avoid implying that an alert automatically represents a diagnosis; present it as a monitored condition requiring appropriate human interpretation.
- The journey is derived from the screens and should be validated against real operational procedures.

Interface principles visible in the system

The design language balances high information density with a recognizable status hierarchy.

Signal hierarchy

Orange highlights alerts and selected/critical states; green communicates neutral room/map tiles; navy anchors navigation and identity.

Consistent sensor language

Repeated icons represent temperature, humidity, ambience, position, door, noise, and SOS across dashboard and device views.

Progressive detail

Summary cards provide a quick scan; detail panels and configuration forms to check connected devices and expose more metadata when needed.

Operational consistency

Repeated table headers, filters, tabs, buttons, and action icons establish familiar patterns across workspaces.

- Design review focus: ensure color is never the only carrier of meaning—pair it with labels, icons, and clear state text.
- Keep typography, icon alignment, spacing, and interaction feedback consistent across the web and mobile apps.
- Protect legibility when the interface is dense, especially in alert cards and wide device tables.

Web app: monitoring dashboard

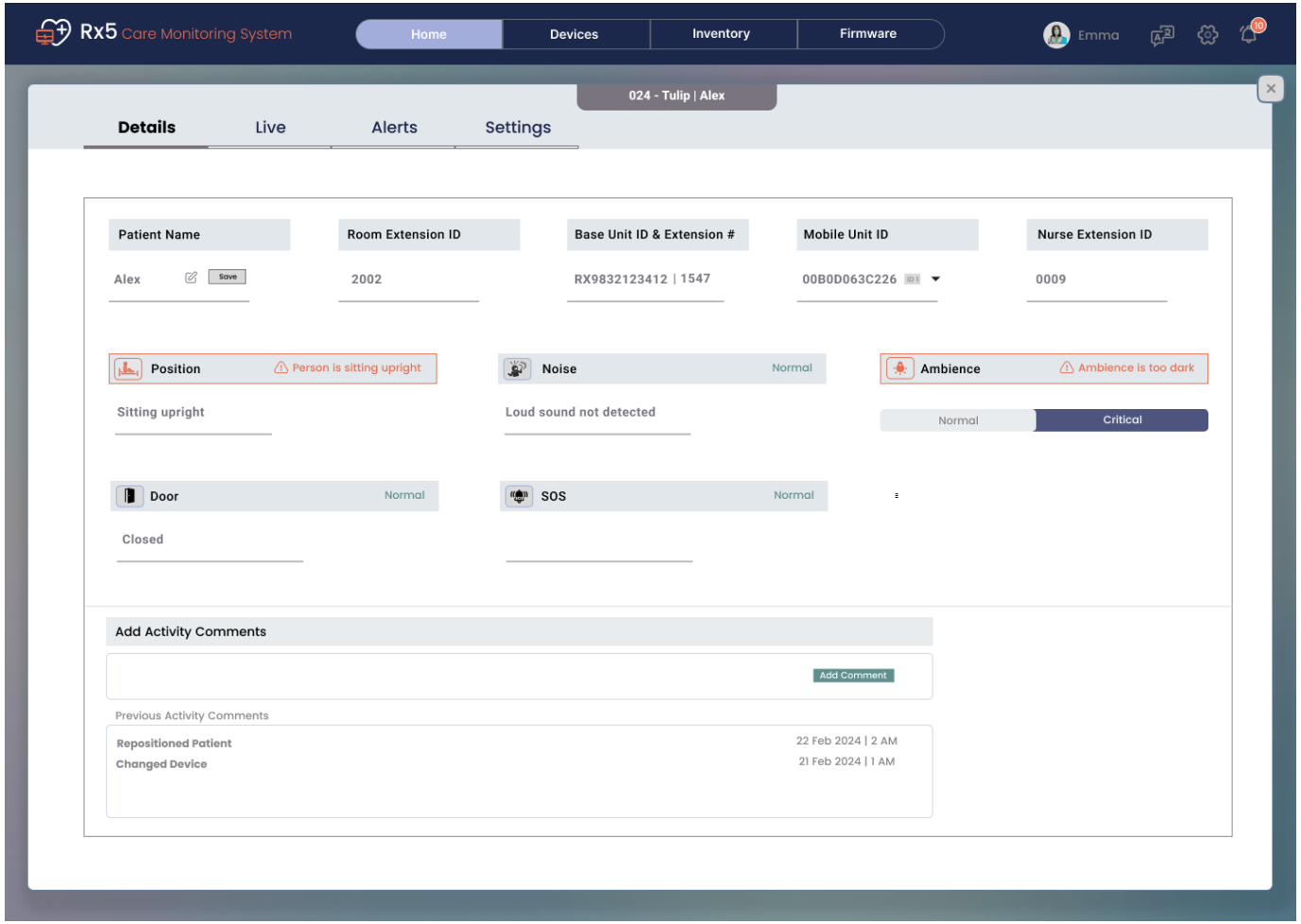
Screen highlights • operational overview



- Global navigation separates Home, Devices, Inventory, and Firmware.
- Quick filters group temperature, humidity, ambience, position, door-open, and noise signals; Refresh All and custom search support scanning.
- Alert cards pair patient/room identifiers with sensor indicators and plain-language messages such as “Person is out of bed” or “Loud sound detected.”
- Battery status, read-out-loud, and live-preview controls sit above the alert area.
- The building map gives room-level context, using distinct tile colors to differentiate selected/alert-associated rooms from neutral rooms.
- Design consideration: provide a clear priority order for multiple simultaneous alerts and explain what each color/state means.

Web app: patient more detail

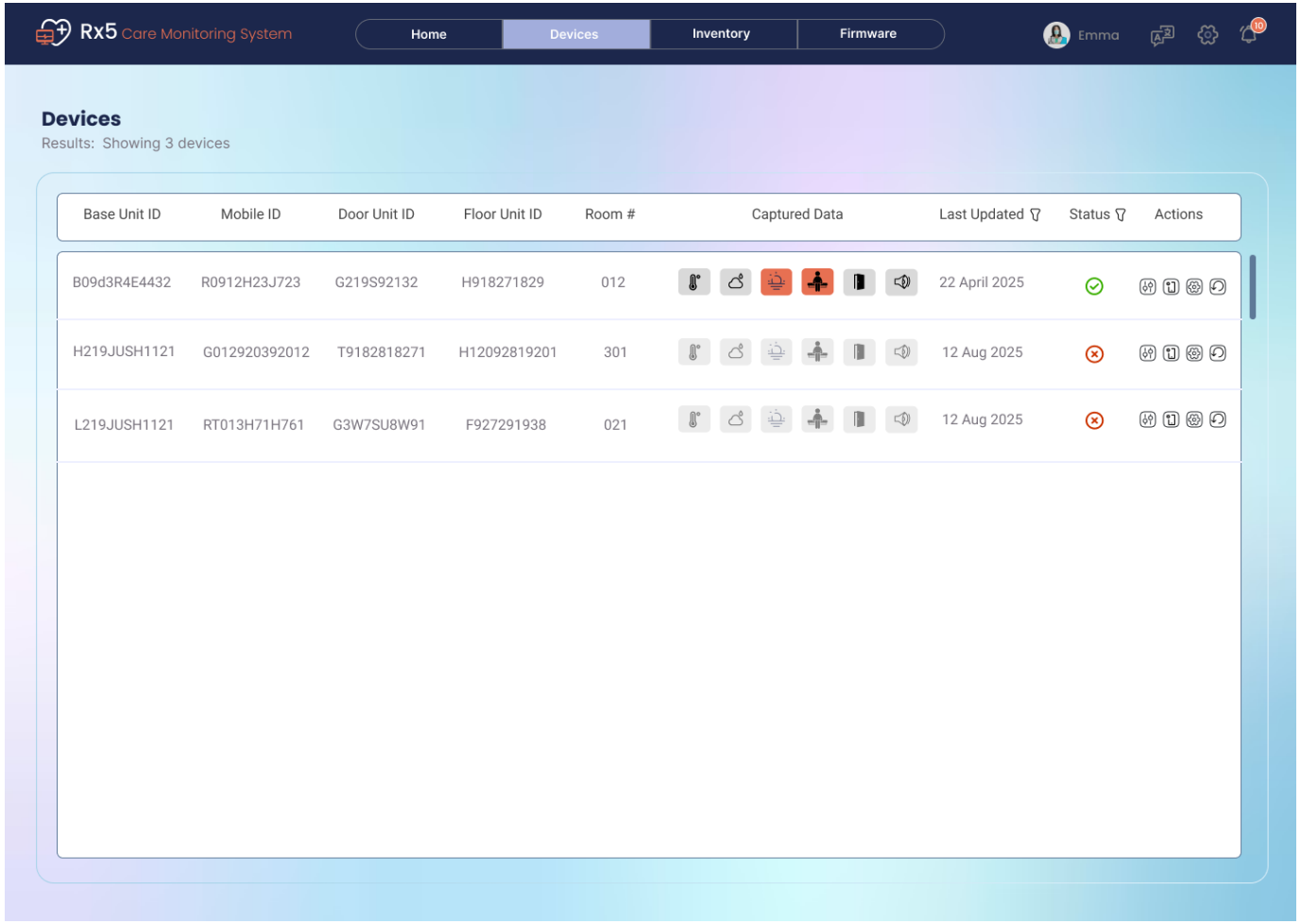
Screen highlights • More details on alerts, connectivity and settings configuration



- Patient detail opens as a focused panel with Details, Live, Alerts, and Settings tabs.
- Identity and linkage fields connect patient name, room extension, base unit ID/extension, mobile unit ID, and nurse extension ID.
- Signal modules expose position, noise, ambience, door, and SOS status; the ambience module visibly distinguishes normal from critical.
- Activity comments support operational context, including a field to add a comment and a previous-comments history.
- Design value: move from a high-level alert to a contextual record without losing the relationship between patient and connected hardware.
- Validation focus: verify status freshness, timestamp clarity, and whether critical states are distinguishable for users with color-vision differences.

Web app: device registry

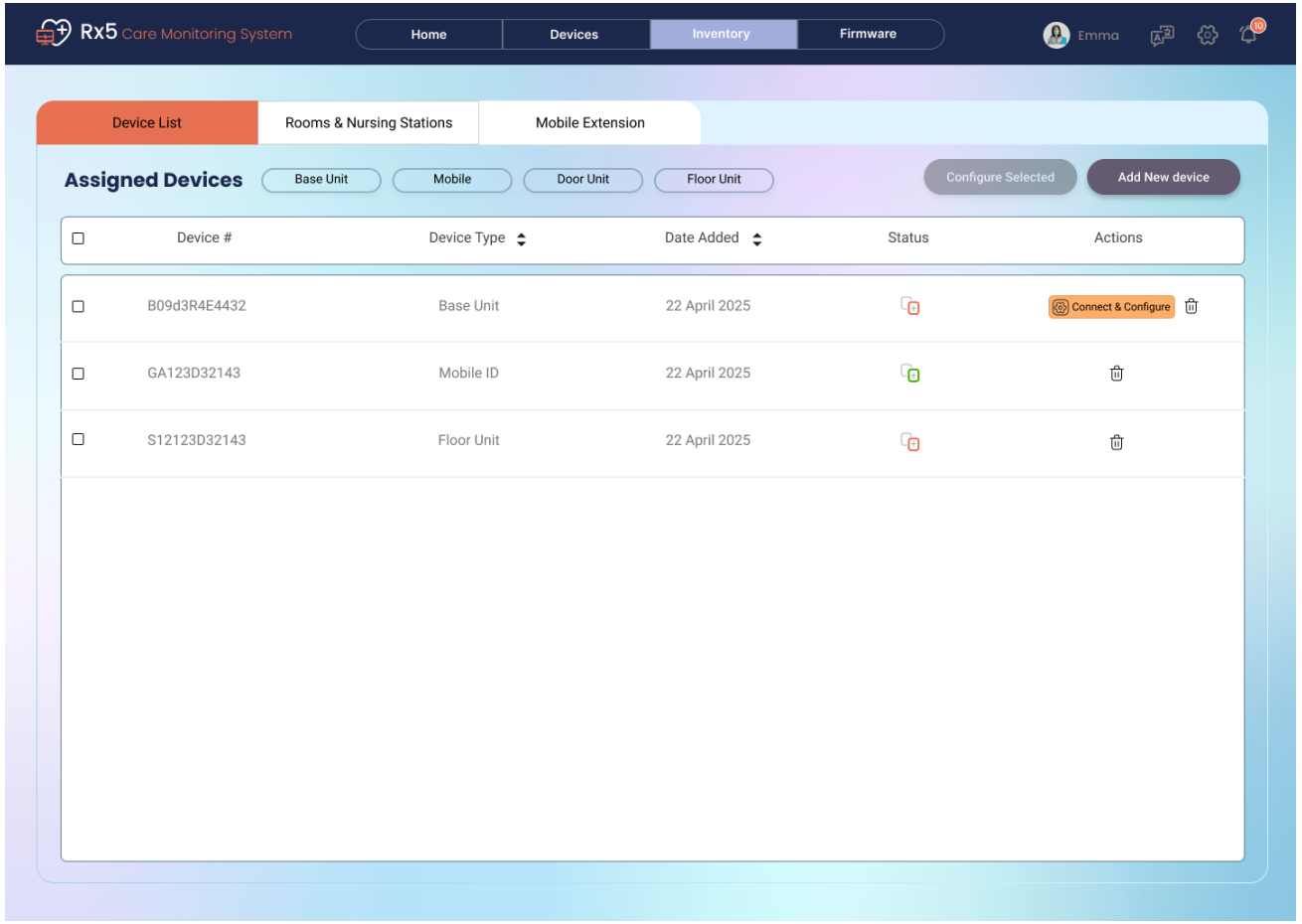
Screen highlights • status, traceability, and action



- The registry aligns base unit, mobile, door unit, floor unit, room number, captured data, last updated, status, and actions in one table.
- Captured-data icons communicate which monitoring capabilities are associated with a device record.
- Last-updated timestamps make recency visible; status icons distinguish successful/available records from exceptions.
- Row-level actions support device-related follow-up without requiring a separate search for the record.
- Design consideration: provide accessible text labels/tooltips for icon-only actions and explicit explanations for failed or stale statuses.
- Data tables should preserve column alignment, sorting/filtering affordances, and predictable row spacing as records scale.

Web app: inventory and device configuration

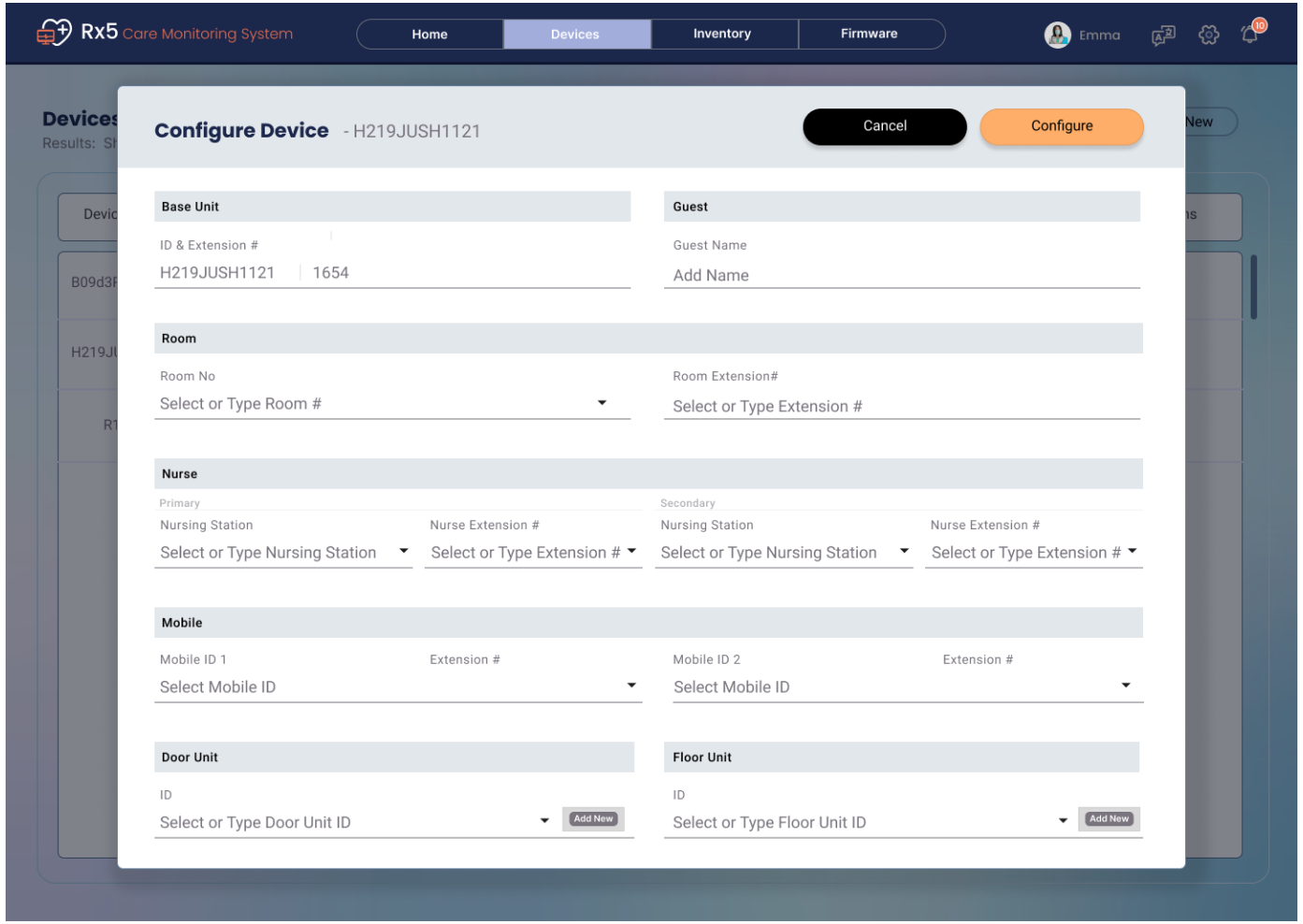
Screen highlights • status, traceability, and action



Manage Rooms and Extensions: Assign or modify device associations with patient rooms and extensions as needed.
Manage Device Lists: View, add, configure, and update base units, door units, floor units, and mobile devices.
Monitor Inventory: Review device assignments, configurations, and statuses to ensure the inventory remains accurate.
Troubleshoot Device Issues: Identify devices that are not sending data, verify their assignments and configurations, and make the necessary corrections to restore monitoring.

Web app: device configuration to room

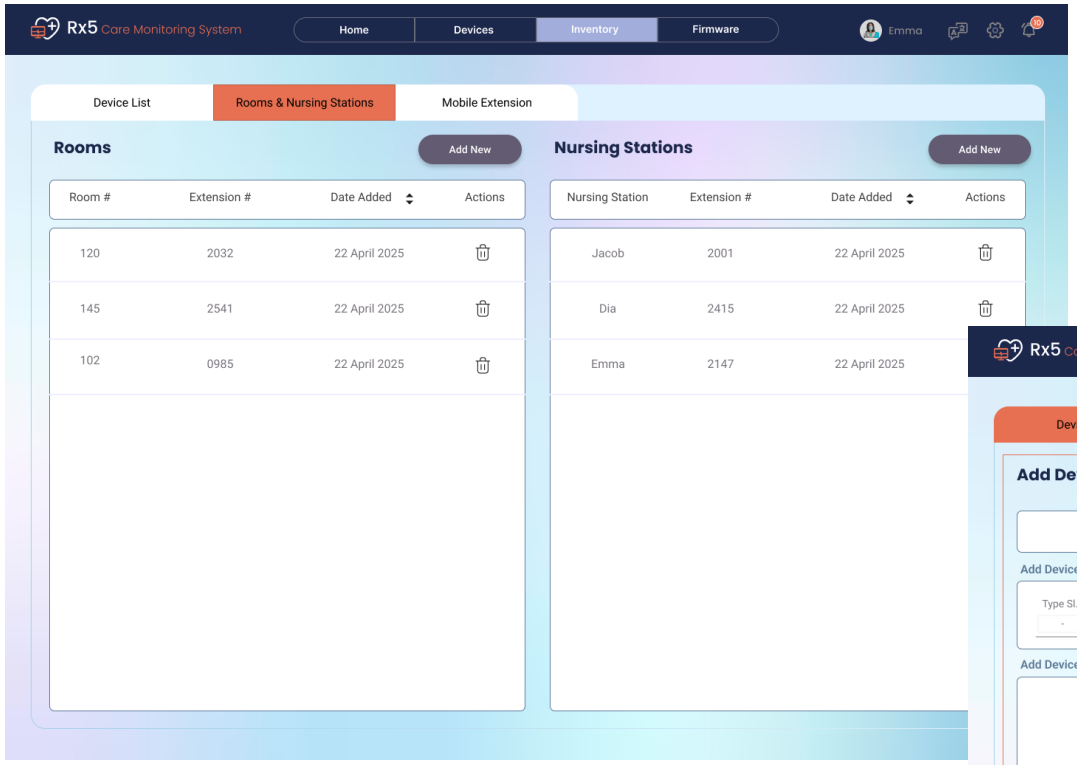
Screen highlights • status, traceability, and action



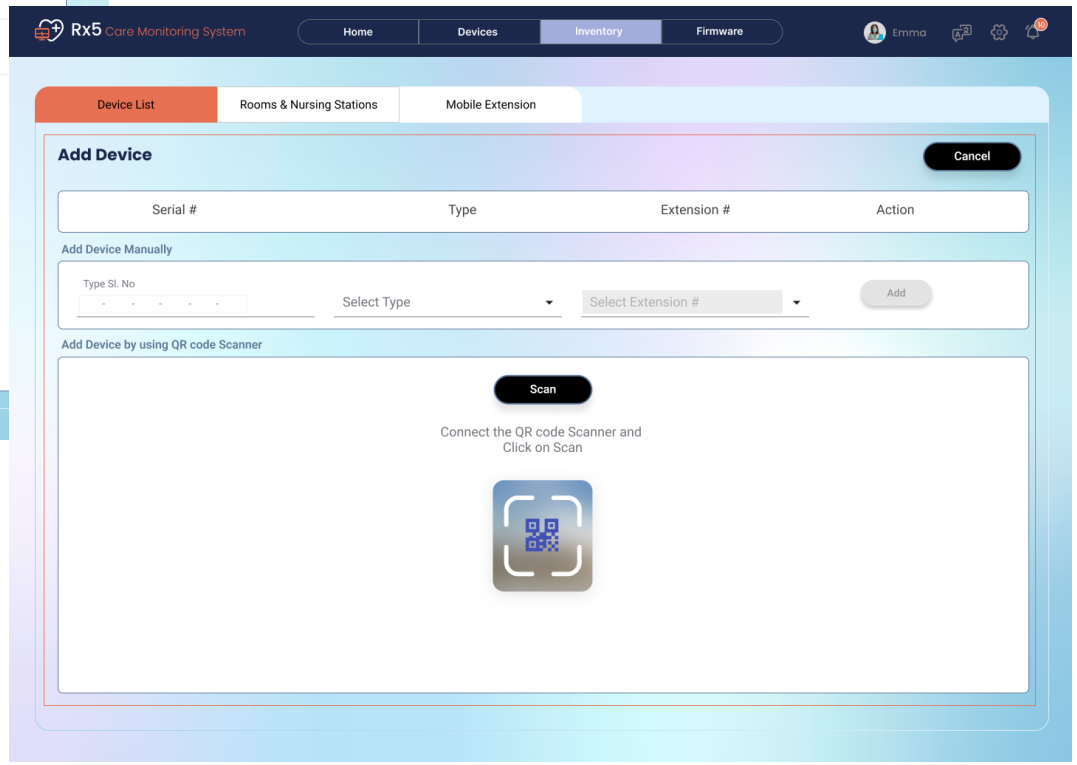
The Configure Device screen allows users to assign and manage the base unit, patient or guest details, room extensions, nursing stations, mobile devices, door units, and floor units.
Configure Base Unit: Verify device ID and extension details and enter the guest name.
Manage Room Assignments: Select or modify room numbers and room extensions.
Assign Nursing Stations: Set primary and secondary nursing stations with their corresponding extensions.
Manage Connected Devices: Assign mobile, door, and floor units, add new devices, and configure the setup to support reliable patient monitoring and alert delivery.

Web app: room administration and device onboarding

Screen highlights • master devices and new device setup

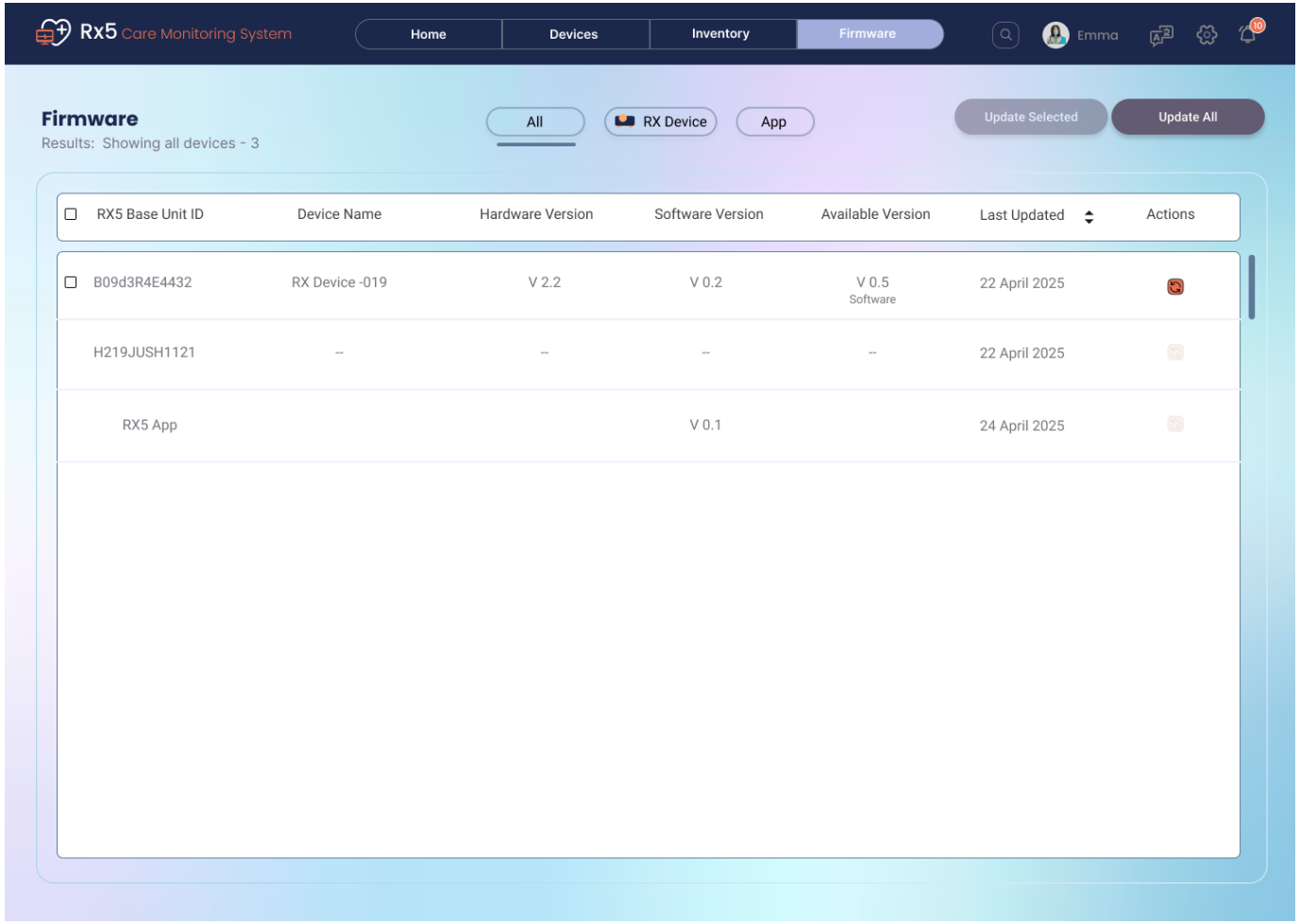


- Rooms and nursing stations are managed side by side, with identifiers, extension numbers, date-added fields, and delete actions.
- The Add New Device view offers manual serial/type/extension entry as well as a QR-code scanning path.
- Providing both manual and scan-based onboarding supports different setup contexts, but the interface should communicate scanner connection state and recovery paths.



Web app: firmware management

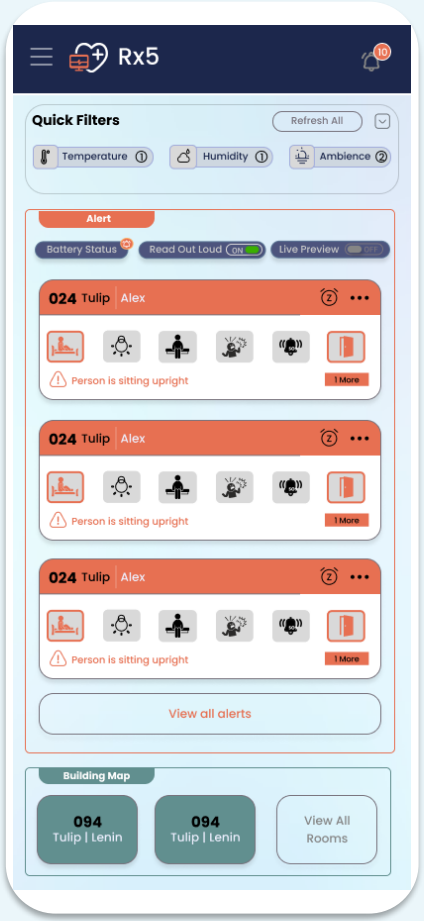
Screen highlights • lifecycle and maintenance



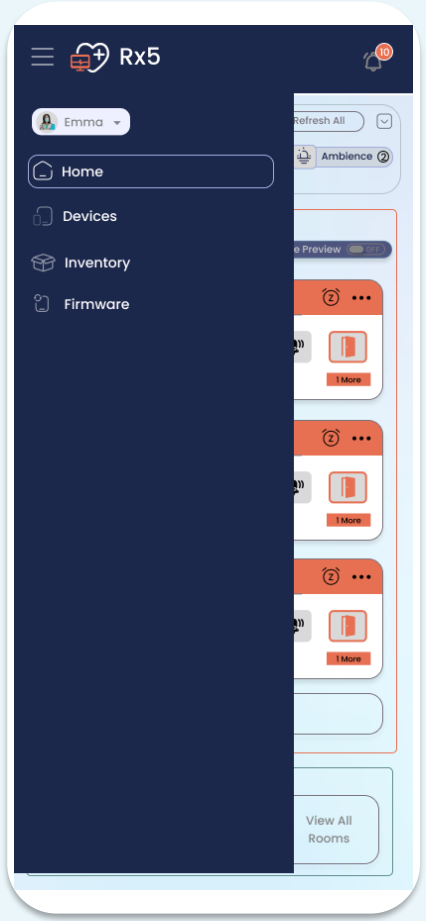
- Firmware workspace filters across all records, RX devices, and the app.
- Rows expose base-unit ID, device name, hardware version, software version, available version, last-updated date, and action.
- Version comparison helps an administrator identify where an update may be available before initiating the update action.
- Bulk actions (“Update Selected” / “Update All”) can improve efficiency but need clear scope, confirmation, progress, and failure recovery.
- Validation focus: clarify compatibility, update eligibility, device connectivity, and the final result of each update.

Mobile app: designed for quick checks on the move

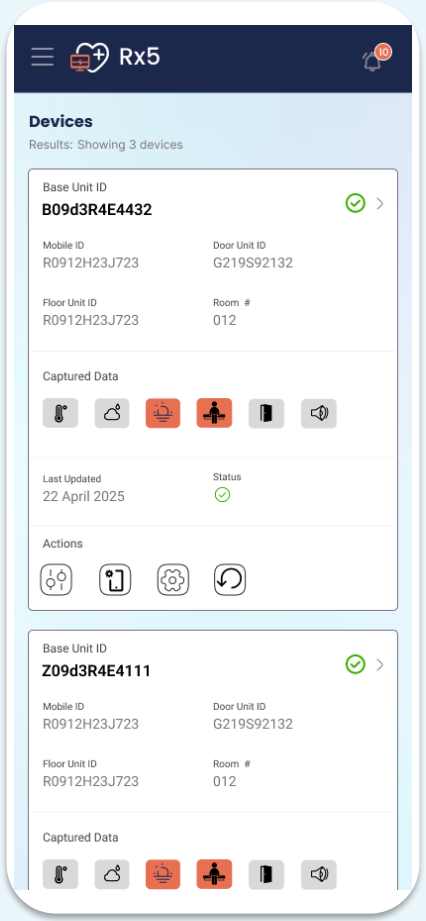
Responsive experience • not simply a compressed web layout



M



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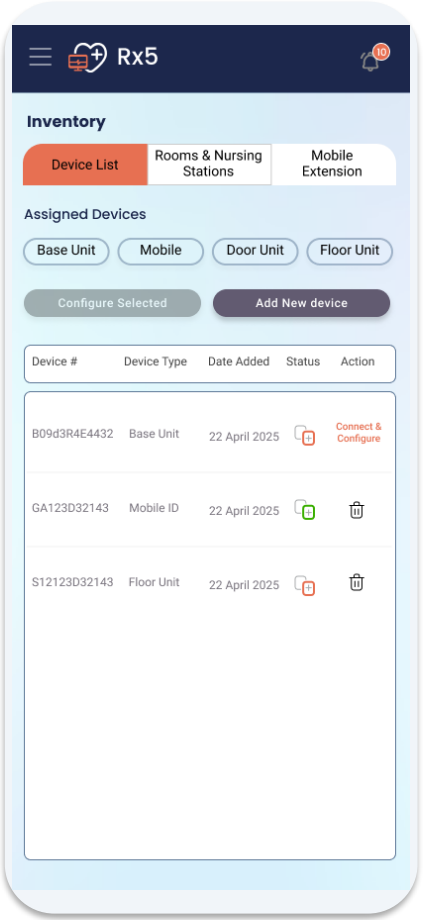


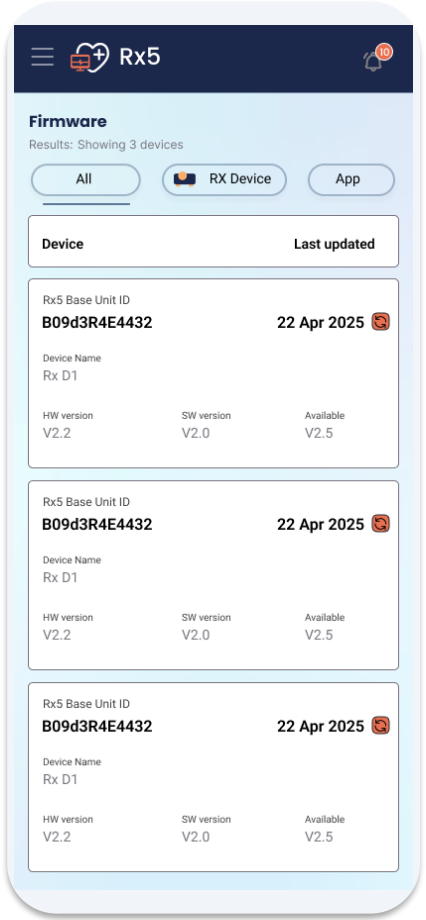
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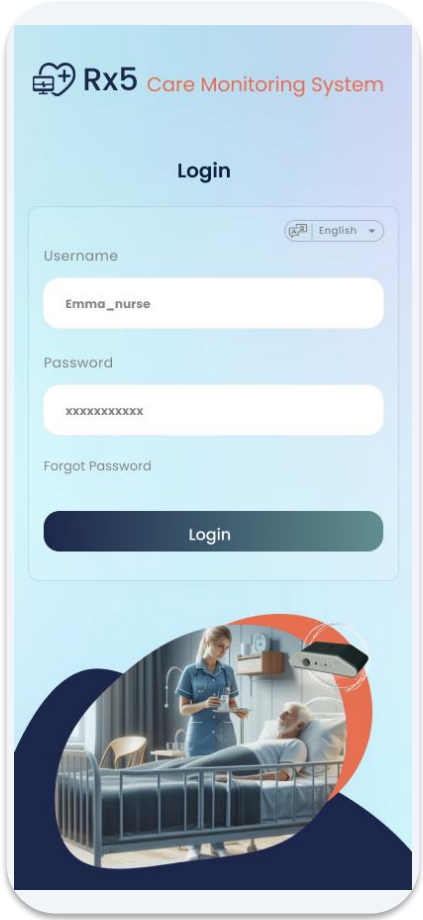
- Quick filters reduce the initial sensor set to the most useful monitoring categories for a compact screen.
- Stacked alert cards keep each patient/room identifier, state icons, warning text, and “more” affordance together.
- The navigation drawer reveals the same four workspaces while preserving a compact top bar and notification indicator.
- Device records become stacked cards so key IDs, captured data, last-updated status, and actions remain readable without wide horizontal tables.

Mobile app: administration at smaller breakpoints

Screen highlights • inventory and firmware







- Inventory converts a dense assignment table into a compact list while preserving device type, date, status, and actions.
- Firmware cards surface device ID, last-updated date, hardware/software versions, and available version in a vertical reading order.
- The sign-in experience maintains the brand identity and core credential flow on mobile.
- Responsive review should check touch-target size, keyboard behavior, scrolling, truncated identifiers, and whether important actions remain discoverable.

Validation plan: prove clarity before claiming impact

Used these checks to validate the design with representative care and technical-administration users.

Alert comprehension

Can users explain the alert state, identify its room/patient context, and distinguish urgent from informational signals?

Findability

Can users locate a device, patient detail, room, nursing station, or firmware record without relying on memory?

Configuration safety

Can users correctly assign units and extensions, recover from invalid inputs, and understand save/cancel outcomes?

Cross-platform continuity

Can users complete the same high-priority tasks on web and mobile, with appropriate expectations at each breakpoint?

- Run moderated task-based usability sessions with realistic scenarios; include accessibility checks for contrast, color independence, keyboard/focus, and touch targets.
- Track task success, time on task, misinterpretations, backtracks, and confidence ratings. Establish a baseline before setting targets.
- Test empty, offline, stale-data, no-permission, update-failed, scanner-unavailable, and multiple-simultaneous-alert states.
- No usability results or quantitative impact metrics were supplied; this is a recommended validation plan, not a claim that testing occurred.

Impact: what the design enables—and what to measure

The screen system establishes a foundation for clearer monitoring and more structured device operations. Product impact should be confirmed with real usage evidence.

Operational visibility

A single dashboard combines alert summaries, sensor filters, and a room-oriented map.

Traceable relationships

Device and patient details expose the IDs and extensions that connect the monitoring ecosystem.

Administrative coverage

Inventory, rooms/nursing stations, onboarding, and firmware are represented as dedicated workflows.

- Measure alert comprehension and time to identify the associated room or patient.
- Measure device setup completion, configuration error rate, and time required to find or update a device.
- Measure firmware update success/failure and recovery time; do not use update availability alone as a success metric.
- Reflection: complex monitoring products need strong state semantics, careful responsive prioritization, and clear recovery paths—not only attractive dashboards.

Roadmap: strengthen trust, accessibility, and resilience

A product-minded follow-through from interface definition to a more robust operational experience.

State model

Define shared semantics for normal, warning, critical, stale, offline, and unknown states across all screens.

Accessibility

Audit contrast, focus order, keyboard navigation, screen-reader labels, icon-only actions, and touch-target sizing.

Resilience

Design loading, empty, partial-connectivity, scanner failure, firmware failure, and permission states.

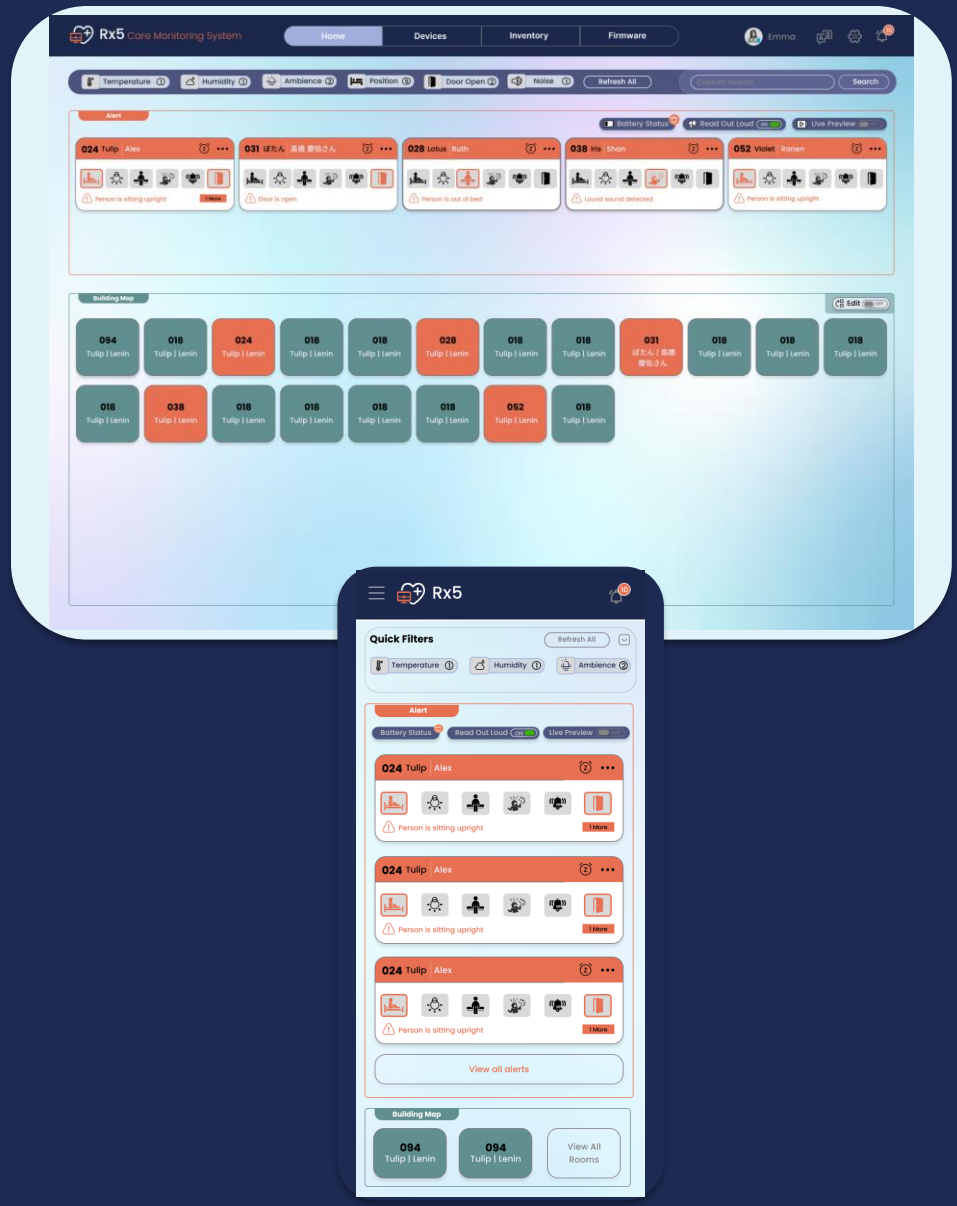
Evidence loop

Run task-based testing, instrument key workflows, review operational feedback, and iterate based on observed friction.

- Align the alert taxonomy and color usage with clinical/operational policies and real-world escalation procedures.
- Document component rules for alert cards, sensor indicators, status badges, tables, filters, forms, and responsive breakpoints.
- Use a release feedback loop to compare intended behavior with how teams actually monitor, configure, and maintain devices.

RX5 CARE MONITORING SYSTEM

Designed for portfolio Showcase



Lead Product Designer • Web + Mobile

A connected care experience should help nurses to care elderly patients in smart ways, understand there needs, and take the next appropriate action.

CASE STUDY • PRODUCT CONTEXT • DISCOVERY • STRATEGY • UX ARCHITECTURE • INTERFACE DESIGN • VALIDATION • REFLECTION